



11.5.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.



Benchmarks

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can determine if a set of triangle side lengths can be used to construct a triangle.
- ☐ I can use the triangle inequality theorem to prove relationships in triangles.



11.6.1 Warm-Up: Number Riddle

1. Use any of the four operations (+, -, $\times$, $\div$) with the numbers shown to produce a result of 36.

7, 0, 2, 3, 5



11.6.2 Exploration: Triangle Inequality Theorem

1. With your partner, use a ruler to determine if the given measurements can be used to create a triangle. Then, complete the table.

Measurements in Centimeters (cm)	Can it create a triangle?	Explanation
3 cm, 6 cm, and 8 cm	☐ Yes ☐ No	
3 cm, 4 cm, and 10 cm	☐ Yes ☐ No	
8 cm, 11 cm, and 15 cm	☐ Yes ☐ No	
2 cm, 5 cm, and 8 cm	☐ Yes ☐ No	
4 cm, 5 cm, and 11 cm	☐ Yes ☐ No	

2. What similarities did you notice about the measurements that could create a triangle?
3. What similarities did you notice about the measurements that could not create a triangle?

4. Make a conjecture about how to determine if side lengths can be used to construct a triangle if you cannot draw it.

5. Ask two other pairs of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of the Conjecture	Initials

6. Review your conjecture with your partner and if necessary, make any changes.

7. Use your conjecture to determine if 3 cm, 4 cm, and 7 cm will create a triangle.

8. Check your answer by constructing the triangle with a ruler.

9. What did you notice about the measurements 3 cm, 4 cm, and 7 cm?

10. Adjust your conjecture if needed.



Triangle Inequality Theorem

The sum of the lengths of any two sides of a triangle is greater than the length of the third side.



If three segments have lengths such that the sum of any two is greater than the third, then a triangle can be constructed from the segments.



11.6.3 Try It: Triangle Inequality Theorem

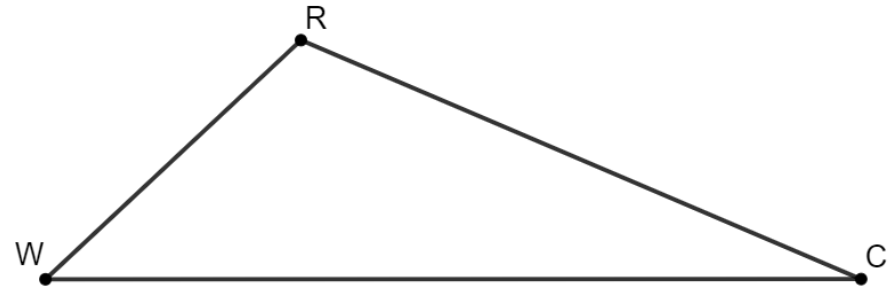
- Complete the table.

Side Lengths	Work	Can it create a triangle?
57, 16, and 75		☐ Yes ☐ No
33, 12, and 25		☐ Yes ☐ No
21, 5, and 26		☐ Yes ☐ No
61, 100, and 61		☐ Yes ☐ No
48, 24, and 21		☐ Yes ☐ No



11.6.4 Exploration: Relating Angle Measures and Sides of Triangles

$\triangle RCW$ is shown, where $WC = 24$, $CR = 18$, and $RW = 10$.



- Use a protractor to find the angle measures. Record your answers in the table.

Angle Measures
$m\angle CWR =$
$m\angle WRC =$
$m\angle RCW =$

- List the side lengths in order from longest to shortest.
- List the angles in order from largest to smallest.

4. What do you notice about the size of the angle compared to the length of the opposite side?



11.6.5 Guided Instruction: Proving Inequality in a Triangle



If two **sides** of a triangle are unequal in measure, then the angle opposite the longer side has the greater measure.



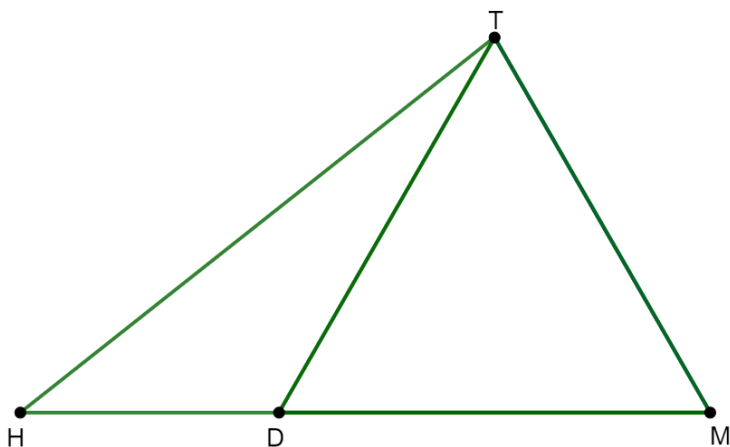
If two **angles** of a triangle are unequal in measure, then the side opposite the angle with the greater measure is the longer side.



Whole Greater Than Parts Theorem

If $a > 0$, $b > 0$, and $a + b = c$, then $c > a$ and $c > b$.

1. $\triangle THM$ is shown, with $\overline{TD}$ creating $\triangle HTD$ and $\triangle DTM$, where $\triangle DTM$ is an equilateral triangle.



Given: $\triangle DTM$ is equilateral; $m\angle DTH = 22^\circ$

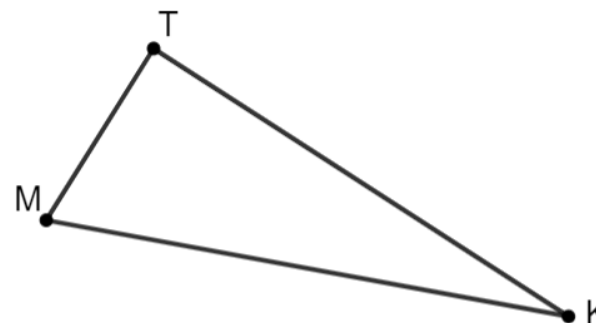
Prove: $HM > HT$

Statement	Reason
1. $\triangle DTM$ is equilateral; $m\angle DTH = 22^\circ$	1. Given
2.	2. Property of equilateral triangles
3. $m\angle DTM + m\angle DTH = m\angle MTH$	3.
4.	4. Substitution Property of Equality
5. $m\angle MTH > m\angle TMD$	5. Whole greater than parts theorem
6. $HM > HT$	6.



11.6.6 Try It: Triangle's Angle Measures

1. $\triangle TKM$ is shown, where $KT = 32$, $MK = 34$, and $TM = 13$. List the angles in order from least to greatest.





11.6.7 Wrap-Up: Cool Down

1. List three side lengths that would not make a triangle. Explain how you know those sides would not make a triangle.
2. List three side lengths that would make a triangle. Explain how you know those sides would make a triangle.



Unit 11, Lesson 7: Centroid of a Triangle



Benchmarks

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.



Learning Targets

- ☐ I can find the centroid of a triangle.
- ☐ I can use the relationship of the distance a centroid is from a vertex to the midpoint of the opposite side to solve problems.